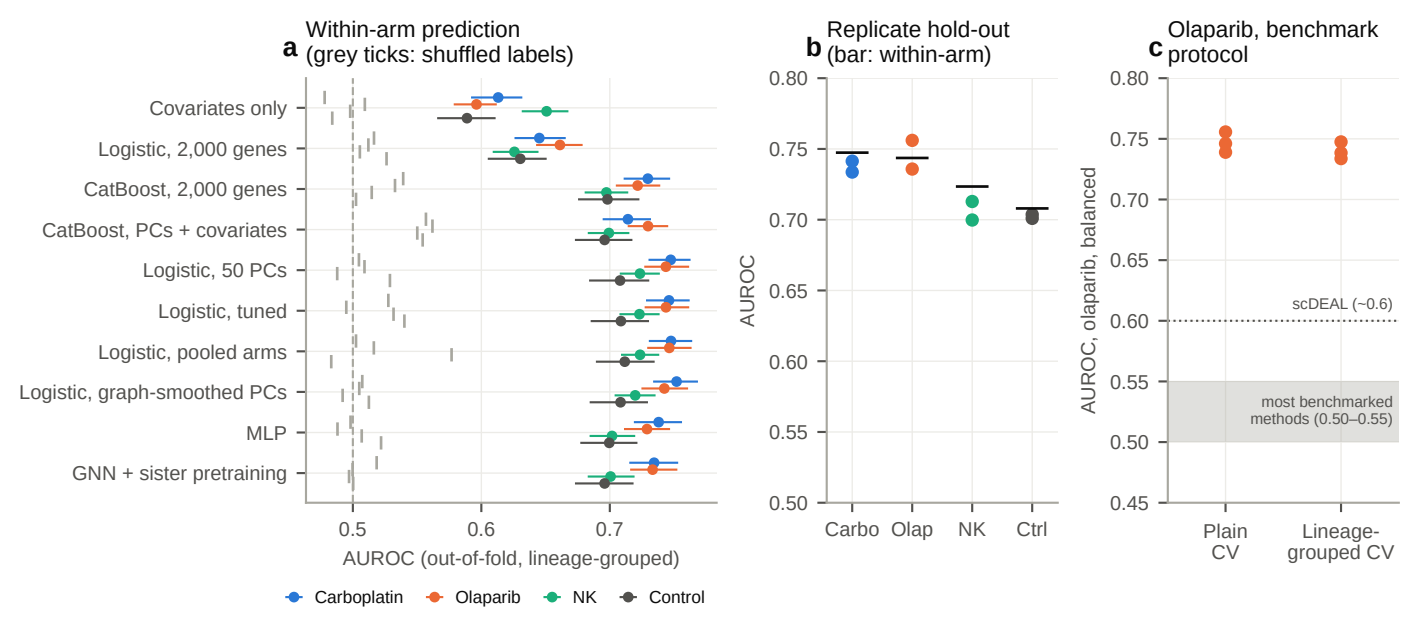
a
Within-arm prediction
(grey ticks: shuffled labels)
Covariates only
Logistic, 2,000 genes
CatBoost, 2,000 genes
CatBoost, PCs + covariates
Logistic, 50 PCs
Logistic, tuned
Logistic, pooled arms
Logistic, graph-smoothed PCs
MLP
GNN + sister pretraining
0.5
0.6
0.7
AUROC (out-of-fold, lineage-grouped)
Carboplatin
Olaparib
NK
Control
b
Replicate hold-out
(bar: within-arm)
AUROC
0.80
0.75
0.70
0.65
0.60
0.55
0.50
Carbo
Olap
NK
Ctrl
c
Olaparib, benchmark
protocol
AUROC, olaparib, balanced
0.80
0.75
0.70
0.65
0.60
0.55
0.50
0.45
scDEAL (~0.6)
most benchmarked
methods (0.50–0.55)
Plain
CV
Lineage-
grouped CV